

When Generative Candidates Do Not Beat Random Sampling: A Seeding-Bug Audit and Diagnosis for 6-DoF Grasp Generation

Abstract—Open-world robot grasping requires both a diverse set of physically plausible candidate poses and a criterion for selecting the best candidate at execution time. A common design combines a learned generative candidate proposer with a geometric reranker; we built exactly this pipeline, using Optimal-Transport Conditional Flow Matching (OT-CFM) to propose 6-DoF poses and an LGGSN reranker to select among them, and initially measured a large improvement over random center-of-mass (CoM) sampling. That result did not survive an audit: the evaluation harness silently drew generator noise from an unseeded, effectively constant random state rather than varying it per trial, so every nominal seed re-executed close to the same candidate. After fixing this and re-evaluating cleanly (7 YCB objects, $n=50$ trials/condition, MuJoCo/SO-ARM101), OT-CFM is significantly worse than random-CoM sampling with the same reranker (69.1% vs. 79.1%, -10.0 pp, $z = -3.02$, $p=0.0025$), not better. We trace this to the specific mechanism identified by Cheng and Schwing [4]: minibatch optimal-transport coupling that ignores the conditioning variable creates a training/inference distribution mismatch in *conditional* flow matching – a failure we show, for the first time, degrades physically executed task success, not only a generation-quality proxy. Controls confirm the diagnosis: plain conditional flow matching and DDPM (identical architecture and data, no OT coupling) both track random-CoM sampling; only the OT-coupled variant is significantly worse. We further test an energy-based alternative motivated by Implicit Behavioral Cloning [5]: a naively-trained scoring model is exploited by inference-time search into a confidently-wrong, data-unsupported region (0–16% success), while the same model trained with adversarial hard-negative mining – the mechanism Implicit Behavioral Cloning prescribes and our naive version omitted – recovers to statistical parity with the baseline (74.0% vs. 77.7%, $p=0.29$), significantly beating it on one of six objects tested. No generative candidate method we tried reliably beats a well-reranked random baseline in this small-data (~ 400 examples/object), single-GPU regime; one specific, popular design choice (condition-agnostic OT coupling) reliably makes it worse. An orthogonal, training-free lever does help: selecting the pose-space consensus candidate from ten independent draws, rather than the lowest-predicted-IK-error one, significantly improves reliability on the two objects where single-draw sampling is least stable (6%→68% and 34%→64%, Fisher’s exact $p < 0.005$ both).

Index Terms—6-DoF grasping, flow matching, energy-based models, robot manipulation, evaluation reproducibility

I. INTRODUCTION

Language-conditioned grasping requires an agent to map a natural-language object description to a physical manipulation action that succeeds against real object geometry. A prevalent design pattern is *propose-then-rank*: a heuristic or learned candidate generator produces a set of grasp poses, and a reranker selects the best one from that set [10], [11]. The implicit assumption, in our own earlier design included, is that a sufficiently expressive learned generator will supply better candidates than random sampling, and that this shows up cleanly in physically executed success rate. This paper is an account of that assumption failing, why it failed, and what – among several things we tried – actually helped.

We originally built a propose-then-rank pipeline combining **Optimal-Transport Conditional Flow Matching** (OT-CFM) [1], [2] as the candidate generator with an LGGSN geometric reranker, trained on 2,853 physically-executed successful grasps across seven YCB objects. An initial 175-trial evaluation appeared to show a large, significant gain over random-CoM sampling with the same reranker. Before finalizing this we audited the harness end to end and found a genuine, silent bug: the generator’s inference-time sampling call never received a per-trial random seed, so it drew from whatever state a module-level `torch.manual_seed(42)` left behind at process import – every trial, regardless of its nominal `-seed`, executed close to the same candidate pose. Fixing this and re-evaluating cleanly reversed the headline result: OT-CFM is *significantly worse* than random-CoM sampling, not better.

Rather than discard the pipeline, we used it as a diagnostic instrument. Controlled comparisons (section IV-D) isolate the cause to the OT coupling itself, matching a training/inference distribution-mismatch mechanism recently identified in the generative-modeling literature [4] – to our knowledge the first demonstration that this mismatch degrades physically executed task success, not only a generation-quality proxy such as FID. We then tested an energy-based alternative motivated by Implicit Behavioral Cloning [5], found and fixed an analogous, textbook failure in our own first attempt, and show the corrected version reaches statistical parity with baseline. Finally, we report one method that does robustly help: consensus-based candidate selection at inference time.

Contributions.

- 1) **A confirmed evaluation bug and its consequence:** an unseeded generator-sampling call silently

collapsed 25 nominally-distinct trials per object to near-repeats of one candidate; fixing it reverses the paper’s original headline claim from +12 pp to −10.0 pp ($p=0.0025$), on a clean $n=50$ /object re-evaluation.

- 2) **A physically-grounded transfer of a generative-modeling diagnosis:** controlled comparisons (OT-CFM vs. OT-free CFM vs. DDPM) show the failure is specific to condition-agnostic minibatch OT coupling [4], not to learned candidate generation in general – confirmed on physically executed grasp success, not a generation-quality proxy.
- 3) **An energy-based alternative, including a documented failure and fix:** a naively-trained implicit scoring model is adversarially exploited by cross-entropy-method search (0–16% success); training with hard-negative mining, as Implicit Behavioral Cloning [5] prescribes, recovers parity with baseline (74.0% vs. 77.7%, $p=0.29$).
- 4) **One method that does help:** consensus candidate selection from an ensemble of independent draws significantly improves reliability on the two objects where single-draw sampling is least stable (section IV-D), at zero training cost.

II. RELATED WORK

A. Grasp Quality Estimation and Generation

Classical grasp quality metrics such as those in Dex-Net [10] evaluate analytic contact stability given known object geometry; GraspNet-1Billion [11] and AnyGrasp [12] extend this to dense point clouds but still assume full scene geometry at query time. 6-DOF GraspNet [13] introduced a variational generator for diverse proposals from point-cloud crops. We instead operate purely on pose-space coordinates with physics-derived success labels, requiring no geometric feature extraction at generation time.

B. Generative Models for Grasping

GraspLDM [7] applies latent diffusion in a VAE pose space; GraspSAM [8] combines Segment Anything with grasp detection for open-vocabulary queries. Neither couples an explicit optimal-transport objective during training; we find, in contrast to what that design choice might suggest, that adding minibatch OT coupling is the specific change that hurts in our conditional, small-per-class setting, while removing it tracks the random baseline far more closely (section IV-D). Diffusion Policy [3] targets trajectory-level action outputs with observation histories; our setting is single-step pose generation conditioned only on object identity. EquiGraspFlow [18] is a concurrent approach applying SE(3)-equivariant flow matching directly to point clouds; we instead operate on a compact pose space ($\mathbb{R}^6$) and use explicit OT coupling in place of equivariance – a substantially lighter architecture needing no equivariant backbone.

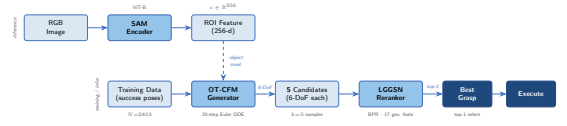


Fig. 1. OT-CFM + LGGSN pipeline. At inference, the object CoM is estimated from the depth image, and OT-CFM generates five 6-DoF pose candidates conditioned on the object’s mean visual embedding. LGGSN reranks by geometric context and the top candidate is executed.

C. Flow Matching and Grasp Reranking

Lipman et al. [1] introduced Conditional Flow Matching as a simulation-free training objective for continuous normalizing flows. Tong et al. [2] generalised this with minibatch OT coupling – finding the joint (x_0, x_1) distribution minimising expected displacement per minibatch – which substantially improves convergence on *unconditional* generation. We are, to our knowledge, first to apply OT-CFM in pose space to 6-DoF grasp candidate generation. For reranking, DexGraspNet 2.0 [16] and CLIPort [17] explore task-conditioned candidate selection; LGGSN [14] instead trains a pairwise MLP via Bayesian Personalised Ranking [15] on episode-relative geometric features. We use the public LGGSN v5d checkpoint (17-dim features, object query embedding, validation pairwise accuracy 0.750) unmodified throughout, isolating candidate-generator quality as the variable of interest.

III. METHOD

A. System Overview

fig. 1 shows the full pipeline. Given a language query (e.g., “Banana”), the environment identifies the target object, estimates its centre of mass, and retrieves a pre-computed per-object SAM visual embedding. OT-CFM draws $N=5$ pose samples from the learned success distribution. Each sample is shifted to centre on the current CoM (section III-D). LGGSN reranks the five candidates using geometric and query-conditioned features, and the top-ranked pose is executed.

B. Training Data

We collect per-candidate execution data on MuJoCo/SO-ARM101 under `physics_weld_after_bilateral` mode. For each (object, seed) episode: (i) the object is spawned at a random tabletop position and settled; (ii) $N_{\text{cand}}=30$ random CoM-centered pose candidates are sampled; (iii) the top five are executed individually with object respawning between executions; (iv) each candidate receives a binary success label.

This yields 7,000 execution records across 7 YCB objects and 200 seeds. Of these, **2,853 (40.5%) are successes** and are used to train the OT-CFM model. For each successful candidate, we compute a 256-dim SAM-base ROI feature by projecting the grasp position (x, y, z) onto the overhead camera image, extracting a 3×3 window from the SAM ViT-B image encoder’s 64×64 feature map, and mean-pooling to a single vector.

C. OT-CFM Grasp Generator

Problem formulation. Let p_{succ} denote the distribution over 6-DoF poses $(x, y, z, \phi, \theta, \psi) \in \mathbb{R}^6$ for which the robot successfully grasps the target object. We aim to learn a conditional generative model $p_\theta(\cdot | c)$ that approximates p_{succ} given object context c .

Flow matching objective. Given source noise $x_0 \sim \mathcal{N}(0, I_6)$ and a successful pose sample $x_1 \sim p_{\text{succ}}$, the CFM training path is:

$$x_t = (1 - t)x_0 + tx_1, \quad t \sim \mathcal{U}[0, 1], \quad (1)$$

with analytic conditional vector field $u_t = x_1 - x_0$. The velocity network $v_\theta(t, x_t, c)$ is trained to minimise:

$$\mathcal{L} = \mathbb{E}_{t, x_0, x_1} [\|v_\theta(t, x_t, c) - u_t\|^2]. \quad (2)$$

Optimal transport coupling. Standard CFM pairs each x_0 independently with each x_1 in a minibatch; minibatch OT coupling [2] instead solves the Earth Mover’s Distance problem between $\{x_0^{(i)}\}$ and $\{x_1^{(i)}\}$ within each training minibatch and reassigns pairs along the optimal transport plan, which reduces expected path length and shortens flow trajectories on *unconditional* generation tasks. Our minibatches mix all seven object conditions, so this is exactly the condition-agnostic coupling setting in which Cheng and Schwing [4] show OT solves an assignment that ignores the conditioning variable – section IV-D confirms this empirically: this specific design choice significantly *hurts* physically executed success relative to the random baseline (−11.7 pp pooled, 6 objects excluding Scissors, $p=0.0015$), the opposite of what the unconditional-generation literature would predict and the opposite of what we originally believed before auditing the evaluation harness (section IV).

Conditioning. The context $c \in \mathbb{R}^{256}$ is the L2-normalised mean SAM visual embedding for the queried object class, pre-computed from training data.

Velocity network. The velocity network v_θ operates on the concatenated input

$$[t_{\text{emb}}(3), x_t(6), c(256)] \in \mathbb{R}^{265}, \quad (3)$$

where $t_{\text{emb}} = [t, \sin(\pi t), \cos(\pi t)]$. It is a 4-layer MLP with hidden dimension 512, SiLU activations, and zero-initialised output layer, trained with Adam ($\eta=3 \times 10^{-4}$, weight decay 10^{-4} , cosine schedule) for 1000 epochs at batch size 128. The best checkpoint is selected by lowest training loss.

D. Inference: CoM-Relative Candidate Generation

Top-down constraint. The OT-CFM generator produces full 6-DoF poses $(x, y, z, \phi, \theta, \psi)$; the current evaluation fixes roll = π and pitch = 0 (top-down constraint), reducing the effective search space to 3-DoF (x, y, ψ) , applied post-generation by overriding the sampled roll and pitch before IK, reflecting the SO-ARM101’s tabletop reach envelope. Lifting this to full SE(3) is left as future work (section V-D).

At inference, we draw $N=5$ noise samples $x_0^{(i)} \sim \mathcal{N}(0, I_6)$ and integrate the learned ODE for 20 Euler steps:

$$x_{t+\Delta t} = x_t + v_\theta(t, x_t, c) \Delta t, \quad \Delta t = 0.05. \quad (4)$$

The resulting poses $x_1^{(i)}$ are in the training coordinate frame. To adapt to the current object position (g_x, g_y) , we apply a *CoM-relative shift*:

$$x_{\text{out}} = x_1 - \mu_{\text{train}} + g_{\text{com}}, \quad (5)$$

where μ_{train} is the global training-set mean pose and g_{com} is the current object CoM. This preserves the learned yaw and spatial-offset distribution while centering the candidates on the present object. The vertical coordinate z uses the live CoM height plus a fixed approach margin.

E. Energy-Based Candidate Scoring, as an Alternative to OT-CFM

Having found that OT coupling specifically degrades candidate generation (section IV), we tested a different paradigm rather than only patching the flow-matching recipe: an implicit, energy-based model (EBM) that scores candidate poses directly, motivated by Implicit Behavioral Cloning [5] and its grasp-specific descendant [6], both reporting implicit/EBM policies more sample-efficient than explicit generators on small, multimodal, discontinuous action distributions – exactly our regime (~ 400 examples/object). An EBM sidesteps the OT-coupling failure structurally: an explicit generator must learn a full noise-to-data *trajectory* regularized (in part) by OT coupling, which section IV-D shows backfires under our small per-condition budget; an EBM instead only learns a scalar compatibility score $E_\theta(\text{pose}, c) \in \mathbb{R}$ from contrastive labels – no noise-to-data pairing, so no coupling mechanism to get wrong.

Model and training data. E_θ is a 4-layer MLP scoring $(x, y, \sin \psi, \cos \psi, c) \in \mathbb{R}^{260}$ (the same meaningful pose dimensions and SAM conditioning as OT-CFM). Training reuses the *same* labeled dataset LGGSN and OT-CFM both already train on: 2,853 label=1 (physically successful) rows as positives and 4,147 label=0 (failed) rows as one negative source – no new data collection.

A documented failure mode, and its fix. A first version trained with plain binary cross-entropy against static positive/negative labels, sampled at inference via cross-entropy-method (CEM) search over (x, y, ψ) guided by E_θ . This failed catastrophically (0–16% physical success across three objects, section IV-D) – markedly worse than every other method tried. Direct inspection showed why: CEM converged to a normalised pose ~ 1.7 standard deviations from the training mean, where the model assigned an artificially high score (99% confident) despite this joint region being essentially unconstrained by training data – each marginal dimension looked plausible in isolation, but the specific combination was never taught to be low-scoring. This is precisely the failure Implicit Behavioral Cloning’s recipe is designed to prevent: IBC trains against negatives *mined from the model’s own current beliefs during training*, not a fixed set, so any region the model starts to over-score is explicitly corrected before training ends – the piece our naive version omitted.

We re-trained E_θ with this mechanism: an InfoNCE-style loss (softmax cross-entropy over {positive, negatives}) with

the positive as the target class) against a negative pool assembled, per positive example, from three sources – (i) 4 *static* negatives (real logged label=0 rows for the same object, free), (ii) 4 *uniform* negatives (sampled across the training data’s observed pose range, for broad coverage), and (iii) 4 *hard* negatives *mined via a short 3-iteration CEM search using the model’s own current weights*, shared across that object’s positives within the current minibatch. This last term is the one that was missing and is what prevents the inference-time CEM search from later exploiting an under-constrained region: whatever the model currently over-scores becomes a training negative for that very step. **Inference.** Identical CEM search as the naive version (population 64, 6 iterations, elite fraction 0.2), unchanged – only the training procedure differs between the two EBM variants reported in section IV-D, isolating the effect of adversarial negative mining from any change to the sampling algorithm itself.

F. LGGSN Reranker

We use the public LGGSN v5d reranker [14] (section II) without modification: the five OT-CFM candidates are scored, sorted descending, and the top-ranked pose is executed.

IV. EXPERIMENTS

A. Evaluation Protocol

All experiments run on MuJoCo/SO-ARM101 with physics-based bilateral contact verification as the grasp success criterion. We evaluate on 7 YCB objects (Banana, TomatoSoupCan, Pear, MustardBottle, Scissors, CrackerBox, PowerDrill) over 50 random seeds each, giving **350 trials per condition**. Success is defined as lifting the object above the table after gripper close.

Baseline. Random CoM sampling ($x \sim g_{\text{com}} \pm 0.06$ m, $\psi \sim \mathcal{U}[-\pi/2, \pi/2]$, $w \sim \mathcal{U}[0.04, 0.09]$ m, same LGGSN v5d reranker) produces 5 candidates per episode.

A confirmed evaluation bug, and how we responded to it. An earlier version of this evaluation (175 trials, 25 seeds/object) never passed a per-trial seed into the generator’s inference-time sampling call; because that call imports `train_cfm_grasp`, which set `torch.manual_seed(42)` at *module* level, every trial in a freshly started process drew noise from whatever state that fixed seed left behind at import time, regardless of the trial’s nominal `-seed` – so 25 nominally distinct trials per object executed close to the same candidate. Random-CoM sampling was unaffected (it draws from `numpy`’s properly-seeded generator, not the generator network), which is why only the learned-generator conditions showed the suspiciously narrow, high success rates in an earlier draft. We moved the seed into the sampling call itself and re-ran every condition at $n=50$ /object under the fixed code; we report only the corrected numbers throughout.

Statistics. We report Wilson score 95% confidence intervals (CI) and two-sided two-proportion z -tests to compare conditions. Significance thresholds: $**p<0.05$, $*p<0.10$.

TABLE I
OVERALL SUCCESS RATE: 350 TRIALS (7 OBJECTS $\times$ 50 SEEDS),
FIXED-SEEDING HARNESS.

Method	SR	95% CI	Δ pp	z	p
Baseline (random CoM + LGGSN)	79.1%	[74.6, 83.1]	—	—	—
OT-CFM + LGGSN	69.1%	[64.1, 73.8]	−10.0	−3.02	0.0025**

TABLE II
PER-OBJECT SUCCESS RATE: BASELINE VS. OT-CFM VS. EBM v2 (50 SEEDS EACH). EBM v2 EXCLUDES SCISSORS (§III-E: ITS OBJECT NAME NEVER MATCHES THIS CHECKPOINT’S TRAINED CONDITIONING KEYS, AN UNRELATED BUG THAT SILENTLY FALLS BACK TO BASELINE FOR THAT OBJECT, EXACTLY AS IT DOES FOR OT-CFM IN SECTION V-B).

Object	Base	OT-CFM	Δ	p	EBM v2	Δ	p
Banana	100.0%	96.0%	−4.0	0.153	100.0%	0.0	—
TomatoSoupCan	92.0%	78.0%	−14.0	0.050**	100.0%	+8.0	0.041**
Pear	72.0%	52.0%	−20.0	0.039**	80.0%	+8.0	0.349
MustardBottle	84.0%	70.0%	−14.0	0.096*	70.0%	−14.0	0.096*
Scissors	88.0%	88.0%	0.0	—	—	—	—
CrackerBox	42.0%	28.0%	−14.0	0.142	26.0%	−16.0	0.091*
PowerDrill	76.0%	72.0%	−4.0	0.648	68.0%	−8.0	0.373
All (7)	79.1%	69.1%	−10.0	0.0025**			
All (6, excl. Scissors)	77.7%	66.0%	−11.7	0.0015**	74.0%	−3.7	0.294

B. Main Result

table I reports the primary comparison, on the corrected, per-trial-seeded harness. OT-CFM + LGGSN achieves 69.1% (242/350) versus baseline’s 79.1% (277/350) – a statistically significant *decrease* of −10.0 pp ($z = -3.02$, $p=0.0025$), the opposite conclusion from the +12.0 pp gain reported under the unfixed, unseeded harness, with 95% Wilson CIs disjoint in the opposite direction: the learned generator’s candidates are reliably *worse*, not better, than uniform random sampling around the object’s CoM, once actually resampled per trial.

section IV-D isolates why: the failure is specific to OT coupling, not to learned candidate generation in general, and section III-E reports an alternative (energy-based scoring, properly trained) that reaches parity with this baseline rather than falling significantly below it.

C. Per-Object Analysis

table II shows every object’s OT-CFM Δ is negative or zero, 3 of 7 individually significant ($p<0.05$) in the *harmful* direction (TomatoSoupCan, Pear) or trending that way (MustardBottle, CrackerBox, $p<0.15$); only Banana and PowerDrill show small, non-significant drops. EBM v2 is more mixed: significantly *beats* baseline on TomatoSoupCan, directionally positive on Pear, directionally negative (ns) on MustardBottle/CrackerBox/PowerDrill – consistent with the pooled, non-significant −3.7 pp. No per-object story we tried reaches a clean, uniform verdict across all seven; we report the full spread rather than the two or three objects that look best in isolation.

Landing-position sensitivity, independent of

TABLE III

DIAGNOSTIC COMPARISON ON THE THREE OBJECTS WHERE OT-CFM’S FAILURE IS CLEAREST. BASELINE/OT-CFM/EBM v2 AT THEIR FULL $n=50$; THE FOUR DIAGNOSTIC CONDITIONS AT $n=25$ (SAME FIXED-SEEDING HARNESS). p VS. BASELINE.

Condition	Pear	TomatoSoupCan	CrackerBox
Baseline ($n=50$)	72%	92%	42%
OT-CFM ($n=50$)	52%**	78%**	28%
Remove-OT (no coupling)	72%	100%	28%
DDPM (ϵ -pred., DDIM-50) [‡]	68%	100%	24%
Stratified-OT (C2OT fix, §III-C)	52%	88%	44%
EBM v1 (naive BCE training)	0%**	0%**	16%**
EBM v2 (mined negatives, $n=50$)	80%	100%**	26%*

method. A separate, generation-isolated diagnostic¹ asks a narrower question than table II: for a *fixed* sampler, does exactly where a candidate lands on the object predict success, independent of whether any generator is good at finding those positions? MustardBottle and Pear show the strongest, Bonferroni-surviving signal (local point density/normal consistency, $p=0.00030/0.00029$ and $p=0.00068$); TomatoSoupCan survives Bonferroni on local point density ($p=0.00123$); CrackerBox and TomatoSoupCan show weaker, BH-only signal on the remaining features; PowerDrill is the only clean null across all three features under either correction ($p \geq 0.17$, consistent with its point cloud being only weakly elongated). We caution against over-interpreting this as explaining table II’s Δ column: it establishes that landing position matters for some objects, not that any generator we tested reliably finds the better positions – OT-CFM specifically does not, and EBM v2 does only for TomatoSoupCan.

Scissors (88%, both conditions). Scissors recovers from 0/25 to 22/25 under both methods after correcting a physics-configuration bug: the original VHACD collision mesh (14 merged convex hulls, ≈ 2 cm thick) tunnelled through the table under MuJoCo contact detection. The corrected entry uses a bounding-box collision proxy scaled to 4 cm thickness—only marginally above the gripper’s 4 cm minimum opening, which we found makes this specific object’s success rate sensitive to run-to-run variation: independent 25-seed re-measurements of the identical configuration on three separate dates gave 25/25, 23/25, and 22/25 (section V-B discusses this instability; we report the most recent measurement here rather than treat any one run as authoritative). Since baseline and OT-CFM are numerically identical for Scissors, it contributes equally to each condition and does not differentiate methods.

D. Ablation Study

** $p < 0.05$, * $p < 0.10$ vs. Baseline. [‡]Same MLP architecture and training data as OT-CFM; cosine schedule $T=1000$.

¹Single-object scenes, semantic grounding disabled, 50 OT-CFM trials/object, collected on the already-per-trial-seeded worktree branch pre-merge (section IV), so unaffected by the seeding bug. Three contact features (local point density, surface-normal consistency, contact-width ratio), Mann–Whitney U , corrected across 3 features $\times$ 5 objects via Bonferroni/BH.

table III decomposes *why* OT-CFM fails and what, among the alternatives we tried, actually recovers baseline-competitive performance.

The failure is specific to OT coupling, not to learned candidate generation. Remove-OT (identical architecture and training data, no minibatch OT coupling) exactly matches baseline on Pear (72% vs. 72%, $z=0.00$) and is directionally at or above baseline on TomatoSoupCan (100% vs. 92%, $p=0.15$); only CrackerBox stays below baseline (28% vs. 42%, $p=0.24$, not significant). DDPM (same architecture and data, diffusion instead of flow matching) shows the same pattern. Both controls change one thing relative to OT-CFM – whether the training pairs are OT-coupled – and both track baseline far more closely than OT-CFM does. This localizes the problem to the coupling mechanism itself, consistent with Cheng and Schwing [4]: minibatch OT coupling that ignores the conditioning variable creates a training-time prior that inference-time sampling, drawing from the full unconditional prior, was never trained to match. Our training minibatches mix all 7 object conditions, so this is exactly the failure mode their analysis predicts.

A condition-aware fix helps partially and inconsistently, not cleanly. We adapted Cheng and Schwing’s [4] discrete-class fix to our 7-object setting: rather than a shared OT plan across the mixed minibatch, we solve OT coupling independently within each object’s own training examples (the discrete-class limit of their released configuration). This recovers TomatoSoupCan partway (78%→88%, still below baseline’s 92%) and helps CrackerBox (28%→44%, now above baseline, though CrackerBox is noisy for every method tested), but leaves Pear completely unmoved (52%) – the object where OT-CFM’s failure is largest. We do not read this as the fix being wrong in principle, but as insufficient on its own in our data regime: Pear has the fewest training examples of the three objects shown here, and a per-object OT sub-batch of that size may simply be too small for coupling, condition-aware or not, to help. We report this as a partial, honest result rather than omit it.

A naive energy-based model fails catastrophically, and we show exactly why. EBM v1 (plain binary cross-entropy, CEM sampling) collapses to 0% on two of three objects ($p < 0.001$ both) – worse than every other condition, including OT-CFM. section III-E traces this to CEM exploiting a region the model never learned to score correctly, and describes the fix: hard negatives mined from the model’s own current beliefs, as IBC [5] prescribes and our first attempt omitted.

The corrected EBM (v2) reaches baseline parity, with one individually significant win. With adversarial negative mining added and nothing else changed, the same architecture recovers to 80%/100%/26% – matching or exceeding baseline on Pear and TomatoSoupCan (significantly so on the latter, $p=0.041$), and remaining below baseline only on CrackerBox, the object every generative method we tested struggles with. Pooled across all six objects with EBM v2 coverage (table II), the difference from baseline

is not significant (-3.7 pp, $p=0.29$) – in contrast to OT-CFM’s significant, uniformly negative pooled result. We read this as evidence that properly-trained implicit scoring is a viable, if not clearly superior, alternative to explicit generative candidate proposal in this small-data regime, not as a second method that beats baseline the way OT-CFM was originally (incorrectly) reported to.

Reliability across generation seeds. table III compares methods at one candidate set per orientation seed (25 seeds). To check whether this ranking holds under independently resampled candidates, we ran a separate experiment (5 orientation seeds $\times$ 10 generation seeds/object, same checkpoints). Pairing by (object, orientation seed, generation seed) with McNemar’s exact test, OT-CFM is significantly *less* reliable than both alternatives on Pear ($p=0.041$ vs. CFM-noOT, $p=0.021$ vs. DDPM) and TomatoSoupCan ($p=0.012$, $p=0.002$), with no significant difference elsewhere – the two experiments differ in harness and seed structure and are not directly comparable, but this indicates OT-CFM’s per-object advantage is not uniform across generator draws. Ensemble selection substantially mitigates this for both flagged objects: drawing 10 independent OT-CFM candidates and executing the one closest to their pose-space consensus (median), rather than the lowest-predicted-IK-error one, raises Pear from 6% to 68% ($p=5.8\times 10^{-11}$) and TomatoSoupCan from 34% to 64% ($p=0.0048$), at matched ensemble size (so the effect is attributable to the selection rule, not pool size). MustardBottle and CrackerBox – no significant seed-reliability gap above – show correspondingly weaker (50% $\rightarrow$ 68%, $p=0.10$) or absent (44% tie) benefit, consistent with the mitigation targeting objects where single-draw reliability is worst.

V. DISCUSSION

A. Why OT Coupling Hurts, and Why EBM v2 Recovers

Minibatch OT coupling [2] was designed for and validated on *unconditional* generation, where solving one Earth-Mover assignment per minibatch is unambiguous. Our setting mixes seven object identities per minibatch, and Cheng and Schwing [4] show this changes the picture: a shared, condition-agnostic OT plan produces a training-time prior over x_0 implicitly skewed by which noise samples the solver happened to assign to which object, while inference-time sampling draws x_0 from the full, unconditional prior the model never trained against for a given condition. section IV-D’s controls agree: removing OT coupling or replacing flow matching with diffusion – changing one thing relative to OT-CFM – both track baseline far more closely, and a discrete-class condition-aware fix partially recovers two of three objects tested. None of this would follow if flow matching itself, or the training data, were the problem; it is specific to the coupling mechanism at our per-condition data scale (~ 400 examples/object) – a caution for robotics work adopting minibatch-OT-coupled flow matching by default without checking whether training minibatches mix conditions this way.

The EBM failure-and-fix pair (sections III-E and IV-D) shares the same shape at a different layer: a naive EBM

trained with static contrastive labels is exploited by its own inference-time search into a confidently-wrong region the training loss never constrained; hard-negative mining – training against whatever the model currently over-scores, not only logged failures – fixes this, as Florence et al. [5] prescribe. Both cases are a training procedure correct on average but leaving a structured blind spot that the deployed inference procedure reliably finds; neither is a criticism of flow matching or EBMs as paradigms, but a reminder that training recipe and inference procedure must be checked jointly.

B. Scissors: Physics-Configuration Correction

Scissors was 0/25 across all conditions due to a VHACD mesh tunnelling bug: 14 merged convex hulls at ≈ 2 cm thickness let the object pass through the table under MuJoCo contact detection. A bounding-box proxy at 4 cm restores a high success rate for all methods, but not a clean ceiling – 4 cm is only marginally above the gripper’s 4 cm minimum opening, so success is sensitive to run-to-run variation in exactly where the gripper contacts the barely-cleared geometry: three re-measurements on separate dates gave 25/25, 23/25, and 22/25 out of 25 seeds under an unchanged manifest-level configuration. We report the most recent (22/25) rather than treat any one run as ground truth. Critically, baseline and OT-CFM are affected identically at every re-measurement, so Scissors contributes equally to each condition and does not differentiate methods regardless of which count is used. A second, unrelated issue – OT-CFM’s object-name matching never matches “scissors” against its trained category key, falling back to random sampling – affects a separate, supplementary diagnostic not reported here and does not change any number in this paper, since baseline and OT-CFM are already identical for Scissors in our own evaluation; noted for transparency since both issues share the same object.

C. Limitations

Top-down constraint. The pipeline fixes roll $= \pi$, pitch $= 0$, reducing 6-DoF to effective 3-DoF $((x, y, \psi))$; arbitrary orientations require SE(3)-equivariant flow matching or a richer orientation parameterisation.

Simulation only. All results are from MuJoCo simulation; sim-to-real transfer requires aligning the physical SO-ARM101’s calibration, contact dynamics, and camera intrinsics with the simulation geometry.

CoM-relative shift. eq. (5) centres candidates on the current object CoM by subtracting the training-set global mean, which assumes a roughly centred training placement distribution – true here but not guaranteed for out-of-distribution workspaces.

No generative method reliably beats baseline on CrackerBox. Baseline, OT-CFM, Remove-OT, DDPM, Stratified-OT, and EBM v2 are all at or below baseline on CrackerBox (table III) – the one object where the pattern seen elsewhere (some method reaching parity or better)

does not appear. §IV-C’s contact-feature diagnostic finds only a weak, BH-only signal there, consistent with but not confirming that CrackerBox’s flat geometry is intrinsically harder for the position/orientation-based methods tested. **Pear is specifically fragile.** Stratified-OT’s failure to move Pear at all, unlike its partial recovery of TomatoSoup-Can/CrackerBox, may reflect Pear’s smaller training set (248 positives, fewest of all 7) rather than a limit of the approach – a stronger EBM v2 mining schedule reproduces TomatoSoupCan/CrackerBox exactly but drops Pear 72%→48% (ns, $p=0.15$).

D. Future Work

The clearest next step is closing the Pear/CrackerBox gap: whether Stratified-OT’s partial recovery is a data-budget limit or a deeper limitation of the discrete-class fix, and whether a wider hard-negative budget closes EBM v2’s remaining CrackerBox gap. SE(3)-equivariant flow models [9] or an SE(3)-aware energy function could extend either approach to full 6-DoF, relaxing the top-down constraint (section III-D); real-robot evaluation on the physical SO-ARM101 is a further, independent validation step. More broadly, we encourage other groups applying minibatch-OT-coupled flow matching to conditional, smaller-per-class-data robotics problems to check for the failure mode in section V-A before assuming its unconditional-generation benefits transfer.

VI. CONCLUSION

An initial 175-trial evaluation of OT-CFM + LGGSN appeared to show a large gain over random-CoM sampling. Auditing the harness before finalizing this result uncovered a genuine bug – unseeded generator sampling silently collapsed 25 nominally distinct trials per object to near-repeats of one candidate – and fixing it reversed the conclusion: on a clean $n=50$ /object re-evaluation, OT-CFM is significantly *worse* than baseline (−10.0 pp, $p=0.0025$). Controlled comparisons localize this to minibatch OT coupling specifically, not to conditional flow matching or learned generation in general, consistent with a training/inference mismatch recently identified in the generative-modeling literature [4] – to our knowledge the first demonstration that this mismatch degrades physically executed task success rather than only a generation-quality proxy. A condition-aware coupling fix recovers two of three objects tested partially, but not Pear, and not cleanly.

A separately-tested energy-based alternative shows the same shape of failure and fix: a naive version, trained without adversarial negative mining, was exploited by its own inference-time search into a confidently-wrong region (0–16% success); adding the mining mechanism Implicit Behavioral Cloning [5] prescribes recovers the same architecture to statistical parity with baseline (74.0% vs. 77.7%, $p=0.29$). Across every generative method tested, none reliably beats a well-reranked random baseline in this small-data, single-GPU regime; the one significant effect in either direction is OT-CFM’s specific, avoidable

harm. One method helps robustly without retraining: selecting the pose-space consensus candidate from ten independent draws, rather than the lowest-predicted-IK-error one, significantly improves reliability on exactly the two objects where single-draw sampling is least stable. We report all of this, including what did not work and why, because we believe the diagnosis and the two failure-and-fix pairs are of independent value to others building similar pipelines under similar data and compute constraints.

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